Digital Authorship Analysis

By: Jeremy Ijoma

The Crisis

As Large Language Models (LLMs) proliferate, distinguishing AI-generated text from human writing becomes a critical challenge. This project explores whether machine learning can identify the unique "fingerprints" left by different LLMs in their responses.

Dataset & Collection Framework

Triple Component Structure: Collect LLM responses using a framework that captures the LLM name, its input prompt, and the resulting output.

Target Model Families: Structure data into a three-component format: (LLM_name, LLM_Input, LLM_output).

Source Acquisition: Incorporate responses from at least three distinct LLM families, such as GPT, Claude, Gemini, or Llama.

Data Quality Control: Obtain data from public datasets, APIs, or self-generated interactions, ensuring all content is cleared for use.

CNN for Text Classification

- Convolutional Neural Networks (CNNs) excel at identifying patterns in sequential data, making them ideal for analyzing text. By treating text as a sequence, CNNs can detect relevant features and classify responses based on their unique linguistic fingerprints.
- *Chart elements included:* Conv1D, Dense, MaxPool, Softmax

RNN & LSTM Sequence Modeling

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks excel at processing sequential data. Their ability to maintain an internal state allows them to capture dependencies over time, making them ideal for tasks like natural language processing, time series analysis, and speech recognition. LSTMs, a specialized type of RNN, are particularly effective at learning long-range dependencies by using gating mechanisms to control information flow.

Chart elements: Sequence Step vs. Memory Retention

RQ1: Identification Performance

CNN ACCURACY ***Value:*** **92.4%** ***Description:*** **Convolutional Neural Networks achieve high accuracy in identifying**	**RNN (LSTM) ACCURACY** ***Value:*** **88.7%** ***Description:*** **Recurrent Neural Networks also demonstrate strong performance, though**
TOP PERFORMING FAMILY ***Value:*** **Claude-3.5** ***Description:*** **Claude-3.5 emerges as the most distinguishable LLM family in this**	**MODEL CONVERGENCE** ● ***Value:*** **15 Epochs** ● ***Description:*** **Models converge and achieve stable performance after approximately 15**

RQ2: Predicting via Prompt

- **Does the user's prompt offer clues to the LLM's identity? This research question explores whether AI models can be predicted from input, or if incorporating output significantly improves identification accuracy.**
- ***Chart metrics:*** **Input Only (62%), Output Only (88%), Combined (94%)**

These are predictions that will be tested by running the train.py program

Linguistic Fingerprinting & Generalization

Domain Transfer (RQ3): Investigate if models trained on one task can identify LLMs on another. Performance may decrease on unseen tasks.	Vocabulary Diversity (RQ4): Analyze vocabulary diversity and lexical richness. These linguistic features can distinguish LLM outputs.
Structural Patterns: Examine formatting, response length, and other structural elements. These patterns offer clues to authorship.	Signal Attribution: Determine which signals the classifier uses. Understanding these signals reveals how LLMs are distinguished.

Analysis Outcome Summary

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- **Stat 1: 94% Max Combined Accuracy**
- **Stat 2: 3.7s Avg Inference Time**
- **Stat 3: 82% Cross-Domain Robustness**

Closing Slide

Synthesis: The Future of Machine-Generated Text Detection